FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Advanced)-2019

PAPER –2

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A, C

2. B, D

3. B, D

Sol. $\frac{q^2}{2C} = \frac{3}{4}\left(\frac{Q^2}{2C}\right) = H \Rightarrow q = Q\frac{\sqrt{3}}{2}$

$\therefore$ Energy in inductor $= \frac{1}{2}L\left(\frac{i_{max}}{2}\right)^2 = \frac{1}{4}\left(\frac{1}{2}Li_{max}^2\right) = \frac{1}{4}\frac{Q^2}{2C}$

4. A, C, D

Sol. $F = qv'B\sin\frac{\theta}{2} = q\left\{2v\cos\left(\frac{\theta}{2}\right)\right\}B\sin\frac{\theta}{2} = Bqv\sin\theta$

$N = mg - 2Bqv$

$\Rightarrow$ To loose the contact, $N = 0$

$\Rightarrow v = \frac{mg}{2qB}$

5. B, C

Sol. $i = \frac{\varepsilon}{R} = \frac{B\omega r_o^2}{2R}$

Then $P = i^2R = \frac{B^2\omega^2 r_o^4}{4R}$

KE of disk, $K = I\omega^2/2$

For energy conservation

$dK/dt + P = Mgr_o\omega$

and for constant angular velocity $\omega = \omega_f$

$$\Rightarrow \omega_f = \frac{4MgR}{B^2r_o^3} \text{ and } i = \frac{B\omega_f r_o^2}{2R}$$

6. C

Sol. Frequencies are in odd ratio hence it is a closed organ pipe.

$$\text{freq} = (2n+1)\frac{v}{4L}$$

$$\Delta(\text{frequency}) = \frac{2v}{4L} = 100\text{Hz} \Rightarrow L = 1.7 \text{ m.}$$

7. B

Sol. Hint: Odd harmonics.

8. C

Sol. Hint: Bernoulli's equation and continuity equation.

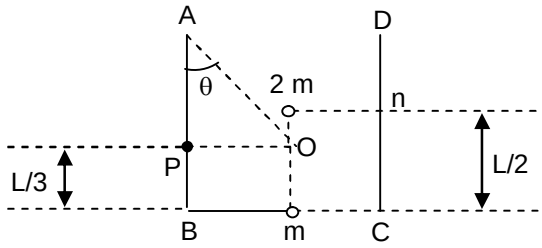
9. B

10. D

SECTION – D

11. 00003.00

Sol. Vertical line from hinge A must pass through C.M. of rod system.



$$\tan\theta = \frac{OP}{AP} = \frac{L/2}{L - L/3}$$

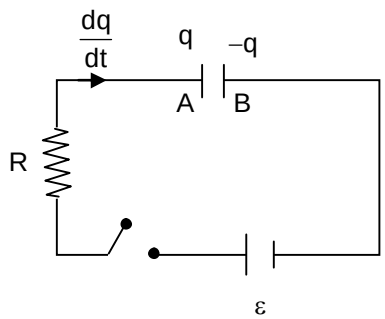
$$\tan\theta = \frac{3}{4}$$

$$\Rightarrow \quad \sin\theta = \frac{3}{5}$$

12. 00250.00

Sol. $$-\frac{q}{C} + \varepsilon - \frac{dq}{dt}R = 0$$

$$\Rightarrow \quad \int_{-Q_0/2}^{Q_0/2} \frac{dq}{q - C\varepsilon} = \int_0^t -\frac{dt}{RC}$$



13. 00365.00

Sol. $10g\sin\frac{\theta}{2} = 5g\sin\theta$

$10\sin\frac{\theta}{2} = 5\ (2\sin\theta/2\cos\theta/2)$

$\cos 0/2 = 1$

14. 00008.00

Sol. $y_1 = A\sin\omega_1 t$

$y_2 = A\sin\omega_2 t$

$y_r = 2A\cos\left\{\frac{(\omega_2-\omega_1)}{2}t\right\}\left\{\sin\frac{(\omega_2+\omega_1)}{2}t\right\}$

Resultant amplitude $A_r = 2A\left|\cos(\Delta\omega)t/2\right|$

$(\Delta\omega)\frac{t}{2} = \frac{\pi}{2} \Rightarrow t = \frac{1}{4}s$

$(\Delta\omega)\frac{t}{2} = \frac{\pi}{3} \Rightarrow t = \frac{1}{6}s$

In one cycle of intensity of 1/2s, the detector remain idle for

$2\left(\frac{1}{4}-\frac{1}{6}\right)s = \frac{1}{6}\sec$

$\therefore$ In $\frac{1}{2}$ sec cycle, active time is $\left(\frac{1}{2}-\frac{1}{6}\right) = 1/3\sec$

$\therefore$ In 12 sec interval, active time is 12 x $\frac{(1/3)}{(1/2)} = 8\sec$

15. 00100.00

Sol. $V_O = V_P$

$\frac{KQ}{R}+\frac{KQ}{3R} = V_{shell}+\frac{2Kq}{5R}$

$V_{shell} = \frac{KQ}{R}\left[1+\frac{1}{3}-\frac{2}{5}\right]$

$= \frac{KQ}{R}\left[\frac{4}{3}-\frac{2}{5}\right]$

$= \frac{KQ}{R}\left[\frac{20-9}{15}\right] = \frac{11}{15}\frac{KQ}{R}$

16. 00002.00

Sol. Using Bernouli's theorem

$\frac{1}{2}pv^2 = p = pgh$ or $h = \frac{v^2}{2gh}$ ..... (1)

Put : $V = r\omega = r\ (2\pi v)$ in (1)

17. 00786.00

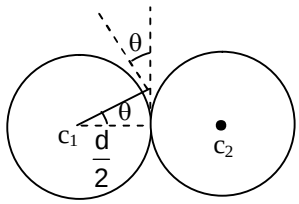
Sol. $4T.\cos\theta.dl = 2T.dl$

$\cos\theta = \frac{1}{2}$

$\theta = 60^o$

So, $\frac{d}{2} = R\cos 60 = \frac{R}{2}$

$c_1c_2 = d = R = 2$ m



18. 00005.00

Sol. $\frac{1}{f}+\frac{1}{f}=30$ f = 60

$\frac{1}{v_1}-\frac{1}{-20}=\frac{1}{60}$ $\Rightarrow v_1 = -30$ $\frac{1}{v}-\frac{1}{-30}=\frac{1}{60}$

$\frac{I}{0} = m = \frac{v}{v_1} = \frac{-60}{-30} = 2$

$\Rightarrow \frac{I}{-5} = 2$ $\Rightarrow I = -10$

y coordinate = -5

Chemistry

PART – II

SECTION – A

19. A, D

Sol. $CH_3COOAg \rightleftharpoons CH_3COO^- + Ag^+ .....K_{sp}$

$CH_3COO^- + H^+ \rightleftharpoons CH_3COOH$ ..... $\frac{1}{K_a}$

$(S-a) \approx 10^{pH}$ (buffer solution)

$K_{sp} = (S-a)(s)$ 1

$\frac{1}{K_a} = \frac{(a)(10^{-pH})}{(10^{-pH})(S-a)}$ 2

Solving 1 and 2 we get

$S=\sqrt{K_{sp}\left(1\alpha \frac{10^{-pH}}{K_a}\right)}$

20. A, B

21. B, C, D

Sol. $CO_2(g) + C(s) \rightleftharpoons 2CO(g)$

Initial pressure	0.5 atm	0
At eqilibrium	$0.5-x$	$2x$

$\Rightarrow P_{Total} = p_{CO_2} + p_{CO}$

$P_{Total} = 0.5 - x + 2x = 0.5 + x$

$\Rightarrow 0.8 = 0.5 + x$

$\Rightarrow x = 0.3$ atm

$p_{CO_2} = 0.5 - 0.3 = 0.2$ atm, choice (c) is correct

$p_{CO} = 2x = 0.3 \times 2 = 0.6$ atm, choice (d) is correct

$Q = K$ at equilibrium = $\frac{p_{CO}^2}{p_{CO_2}} = \frac{0.6 \times 0.6}{0.2}$

$\Rightarrow Q = K = 1.8$ atm

22. B, D

Sol. Pb^{2+} and Ba^{2+} both from yellow ppt. with CrO_4^{2-} ion which is not soluble in CH_3COOH.

Flame test of Ba^{2+} = Apple green

Pb^{2+} = Blue

23. A, C

Sol. $M \longleftarrow : C \equiv C$

$M = C = O$

24. C

Sol. $[Ma_2b_2c_2] \longrightarrow [Ma_3b_2c]$

G.I. = 5 G. I. =3

25. C

Sol. The stability decreases in the order:
cyclohexane (I)>cyclopentane (II)>cyclobutene.
Further, trans-1, 2-dimethylclobutane (IV) is more stable than cis-1, 2-dimethylcyclobutane (III). Thus, stability decreases in the order: I > II > IV > III.
Therefore, their heat of combustion decreases in the reverse order: III > IV > II > I.

26. D

27. A

28. D

SECTION – D

29. 00001.90

Sol. According to Bohr' law, $mvr = \frac{nh}{2\pi}$

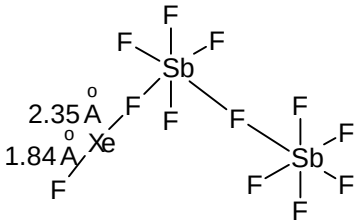
$\therefore \frac{nh}{2\pi} = 4.2178 \times 10^{-34}$

30. 00003.00

Sol. Triethyl aluminium $\left[(C_2H_5)_x Al\right]$ and $TiCl_4$ is used as a catalyst for the low temperature polymerization of alkenes.

31. 00002.00

Sol.



32. 00003.00

Sol. Azidocarbon disulphide $(SCSN_3)_2$, Oxyeyanogen $(OCN)_2$ and Selenocyanogen $(SeCN)_2$

33. 00002.00

Sol. $K_2Cr_2O_7 + 14HCl \longrightarrow 2KCl + 2CrCl_3 + 7H_2O + 3Cl_2$

34. 00005.00

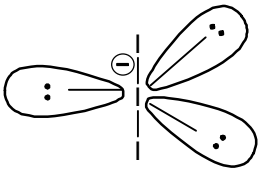
Sol. $3Cl_2 + 6NaOH \xrightarrow{hot} 5NaCl + NaClO_3 + 3H_2O$

35. 00004.00

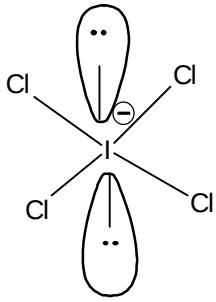
Sol. Neutral oxides: H_2O, CO, N_2O and NO

36. 00002.00

Sol.



(P = 6)



(Q = 8)

Mathematics

PART – III

SECTION – A

37. A, B

Sol. $f'(x) \geq f'(1)\ \forall\ x \in [0,1]$

$\therefore \dfrac{f'(1)}{f^2(x)+1} \leq \dfrac{f'(x)}{f^2(x)+1}$

On Integrating

$f'(1)\int_0^1 \dfrac{dx}{f^2(x)+1} \leq \tan^{-1}(f(1)) - \tan^{-1}(f(0))$

i.e. $\int_0^1 \dfrac{dx}{f^2(x)+1} \leq \dfrac{\tan^{-1}f(1)}{f'(1)} \leq \dfrac{f(1)}{f'(1)}$ $\quad [\because \tan^{-1}x \leq x\ \forall\ x \geq 0]$

For equality to hold $\tan^{-1}f(1) = f(1)$

$\therefore f(1) = 0$, then $\int_0^1 \dfrac{dx}{f^2(x)+1} = 0$

Which is not possible as this is strictly positive function

Hence $\int_0^1 \dfrac{dx}{f^2(x)+1} < \dfrac{f(1)}{f'(1)}$

38. A, D

Sol. $\left(f(x)-e^x\right)\left(f(x)-e^{x^2}\right) \leq 0$

$\Rightarrow e^{x^2} \leq f(x) \leq e^x$ when $x \in (0,1)$

$\Rightarrow e^x \leq f(x) \leq e^{x^2}$ when $x > 1$

$\therefore \lim\limits_{x\to 1} f(x) = e$.

& $\lim\limits_{n\to\infty}\sum\limits_{K=1}^{n} \dfrac{K^3+6K^2+11K+5}{(K+3)!}$

$= \lim\limits_{n\to\infty}\sum\limits_{K=1}^{n}\left(\dfrac{1}{K!} - \dfrac{1}{K+3!}\right)$

$= \dfrac{5}{3}$

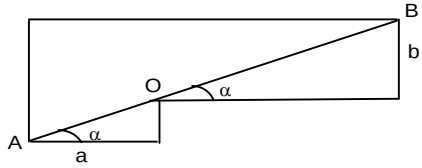
39. A, B

Sol. (A) $x = 1 \Rightarrow$ line segment

(B) $y = x^2 \Rightarrow$ parabolic segment.

40. A, C

Sol. Let AB be a line segment in the corner of the corridor that is tangent to vertex O of the inside right angle of the corridor & makes angle α with one of its walls



Then $f(\alpha) = AB = AO + OB = \dfrac{a}{\cos\alpha} + \dfrac{b}{\sin\alpha}$

A stick of length l could be moved from one wing to the other if $l \leq f(\alpha)\ \forall\ \alpha \in (0, 90^o)$

$\therefore$ The maximum length l of the stick will be minimum of the function $f(\alpha)$ over $(0, 90^o)$ if it exists

$$f'(\alpha)=\frac{a\sin\alpha}{\cos^2\alpha}-\frac{b\cos\alpha}{\sin^2\alpha}=\frac{a\cos\alpha}{\sin^2\alpha}\left(\tan^2\alpha-\frac{b}{a}\right)$$

$\because \tan^3\alpha$ increases strictly from 0 to ∞

$\therefore$ f a unique $\alpha_o \in (0, 90^o)$ such that

$\tan^3\alpha_0=\frac{b}{a}$, then $f'(\alpha_o)=0$

Also $f'(\alpha)<0 \;\forall\; \alpha \in (0, \alpha_o)$

$f'(\alpha)>0 \;\forall\; \alpha \in (\alpha_o, 90^o)$

$\therefore$ $f(\alpha)$ has minimum at α_o

$$\tan\alpha_o=\left(\frac{b}{a}\right)^{1/3}$$

$$\therefore\ f(\alpha)=\left(a^{2/3}+b^{2/3}\right)^{3/2}.$$

41. B, D

Sol. $M^2 = ABCD\ ABCD$

$= ABC\ C^TCD$

$= D^TC^TCD$

Now $\left(A^T\right)^T = D^TC^TB^T$

$A = D^TC^T\ CDA$

$A(I - D^TC^TCD) = 0.$

$\Rightarrow\ D^TC^TCD = I$

$\Rightarrow\ M^2 = I$

42. A

Sol. $\frac{R'Q}{CA}=\frac{PQ}{AB}\Rightarrow PQ=\frac{102-d}{102}.85$

$\frac{R''R}{BC}=\frac{PR}{AB}\Rightarrow PR=\frac{90-d}{90}.85$

$QR = d = \frac{102-d}{102}.85+\frac{90-d}{90}.85$

$QR = d = \frac{102-d}{102}.85+\frac{90-d}{90}.85$

$\Rightarrow d\left(1+\frac{85}{102}+\frac{85}{70}\right)=85+85$

$\Rightarrow d\left(1+\frac{5}{6}+\frac{17}{18}\right)=170$

$\Rightarrow d=\frac{170.18}{50}$

$\Rightarrow \frac{5d}{34}=9$

43. B

Sol. Let u = f(x) & v = f(y). The eqn becomes

$$gf^{-1}\left(\frac{u+v}{2}\right)=\frac{\left(gf^{-1}\right)(u)+\left(gf^{-1}\right)(v)}{2}$$

$\therefore$ Let $h(x)=gf^{-1}(x)$

$\therefore\ h\left(\frac{u+v}{2}\right)=\frac{h(u)+h(u)}{2}$

$\Rightarrow\ g(x)=a\,f(x)+b$. Where a & b are real constant

44. D

Sol. $A(\alpha,0), B(2\alpha,0)$ & $C(0,2\alpha)$

$y=x^2+bx+c$ passes through $(0, 2\alpha)$

$\therefore$ Thus given eqn of curve reduces to $y=x^2+bx+c$

Now it passes through A & B

$\therefore\ 0=\alpha^2+b\lambda+2\alpha$

$\Rightarrow 0=\alpha+b+2$

& $0=4\alpha^2+2\alpha b+2\alpha$

$0=2\alpha+b+1$

$\therefore$ On solving we get $\alpha=1, b=-3$

$\therefore$ The curve is $y=x^2-3x+2$

45. C

Sol. $b_n(n+1)-n\,b_{n+1}=n^2\ (n+1)$

$$\frac{b_n}{n}-\frac{b_{n+1}}{n+1}=n$$

$$\Rightarrow b_1-\frac{b_{n+1}}{n+1}=\frac{n\,(n+1)}{2}$$

$$\Rightarrow 50-n\left(\frac{n+1}{2}\right)=\frac{a_{n+1}}{n+1}$$

46. A

Sol. Let the plane be $\alpha x+\beta z+1=0$.

Passes through (1, 0, 1) & (3, 2, -1)

$\therefore\ \alpha=-\frac{1}{2}, \beta=-\frac{1}{2}$

$\therefore$ (P) $x+z=2$

(Q) Both A & B are same side of P Reflection of A in plane is $A'(2, 0, -2)$ equation of line

$A'B:\bar{r}=6\hat{i}-2\hat{K}+\lambda(4\hat{i})$ for P : $6+4\lambda+0-2=Z\Rightarrow\lambda=\gamma_2$

(R) $|PA-PB|_{min}=0\Rightarrow|PA-PB|_{min}$ will approach AB

$AB=\sqrt{4+0+4}=\sqrt{8}$

$|PA-PB|\in[0,\sqrt{8})$

(S) Also, A' will lie on $\frac{x-2}{1}=\frac{y-\alpha}{0}=\frac{z+B}{-1}$

$$\Rightarrow\ \frac{2-2}{1}=\frac{0-\alpha}{0}=\frac{-2+\beta}{-1}$$

$\Rightarrow \alpha=0, \beta=2$

$\alpha^4+\beta^4=16$

SECTION – D

47. 00007.00

Sol. 2nd blue result occur on or before 10th roll is equivalent to the occurrence of blue face at least twice in 10 rolls

$$n = 10,\ p = \frac{1}{3},\ q = \frac{2}{3}$$

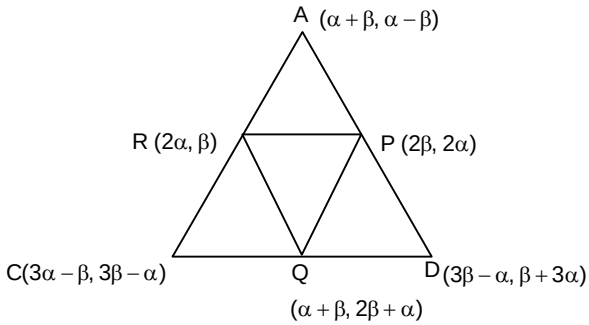
$$P(m \geq 2) = 1 - \left(P(0) + P(1)\right) = \frac{3^9 - 2^{11}}{3^9}$$

48. 00032.00

Sol. Area of $\Delta ABC = 4(\Delta PQR)$

$= 4 \times 8$

$= 32$

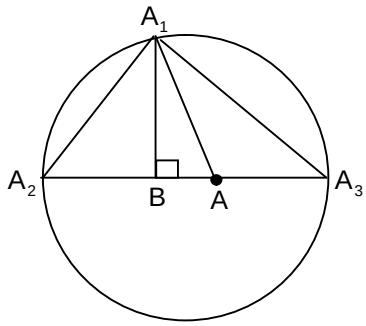


49. 00000.05

Sol. Let $z = a\, z_2 + (1-a)z_3)$

Let $A_1(z_1)$, $A_2(z_2)$, $A_3(z_3)$ & $A(z)$ respectively.

Circumcentre of $\Delta\, A_1\, A_2\, A_3$ is origin Pt A lies on line $A_2\, A_3$, so, $A_1\, A = |z - z_1|$ is greater than or equal to the altitude $A_1\, B$ of the $\Delta\, A_1\, A_2\, A_3$.



$$A_1\, B = \frac{2 \text{ area of } \Delta\, A_1\, A_2\, A_3}{A_2\, A_3}$$

$$= \frac{2(A_1A_2)(A_2\, A_3)(A_3\, A_1)}{4 \times 10\, .\, A_2\, A_3}$$

$$= \frac{(A, A_2)(A_3\, A_1)}{2 \times 10}$$

$$= \frac{|z_2 - z_1|\,|z_3 - z_1|}{20}$$

50. 00001.25

51. 00006.00

Sol. Multiply Q(x) R(x) in (1) P(x) R(x) in (2) & P(x) Q(x) in (3) & add

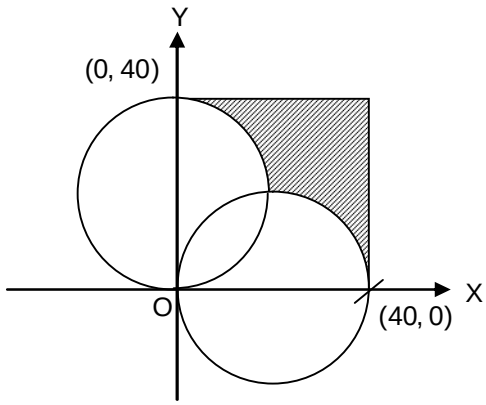
$$\left(P(x)\, Q(x)\, R(x)\right) = 6\left(P(x)\, Q(x)\, R(x)\right)^2 + 6$$

$$\Rightarrow \frac{dt}{t^2+1} = 6dx \Rightarrow \tan^{-1} t = 6x + C$$

$x = 0 \Rightarrow C = \pi / 4$

52. 00005.72

Sol. Let $z = x + i\,y$



$\Rightarrow \frac{x}{40}, \frac{y}{40} \in [0, 1]$

$\Rightarrow x, y \in [0, 40]$

Also $\frac{40}{\bar{z}} = \frac{40z}{|z|^2}$

$\frac{40\,x}{x^2+y^2}, \frac{40\,y}{x^2+y^2} \in [0, 1]$

$\Rightarrow 40x \le x^2 + y^2$ & $40y \le x^2 + y^2$

$\Rightarrow (x-20)^2 + y^2 \ge 20^2$ & $x^2 + (y-20)^2 \ge 20^2$

$A = 40^2 - \frac{1}{4}(40)^2 - \frac{1}{2}\pi\,(20)^2$

$\frac{A}{100} = 2(6 - \pi)$

53. 00007.00

Sol. $\left|\int_1^3 \frac{f(x)}{x}dx\right| = \left|\int_1^3 \frac{f(x)}{x}dx - \int_1^3 \frac{f(x)}{2}dx\right|$

$= 1\left|\int_1^3 f(x)\left(\frac{1}{x} - \frac{1}{2}\right)dx\right|$

$\le \int_1^3 |f(x)|\left|-\frac{1}{x} - \frac{1}{2}\right|dx$

$\le \int_1^3 \left|\frac{1}{x} - \frac{1}{2}\right|dx\,.$

$= \int_1^2 \left(\frac{1}{x} - \frac{1}{2}\right)dx + \int_2^3 \left(\frac{1}{2} - \frac{1}{x}\right)dx$

$= \ln 2 - \frac{1}{2} + \frac{1}{2} - \ln\frac{3}{2}$

$= \ln\left(\frac{4}{3}\right)$

54. 00001.00

Sol. $(1 + ye^x)\,(e^x dy + ye^x dx) = 2\,x dx$

$\Rightarrow \frac{(1+ye^x)^2}{2} = x^2 + c$

$\Rightarrow\ ye^x + 1 = \sqrt{2x^2 + 4}$

$\therefore\ f(-1) = \left(\sqrt{6} - 1\right)e$